



Application Note: AN1200.106

LR2021 Series Crystal Temperature Mitigation

Table Of Contents

1	Introduction.....	4
2	Crystal Thermal Frequency Drift	5
2.1	Origin of the Frequency Drift.....	5
2.2	Impact.....	5
3	PCB Solutions.....	7
3.1	Reduce Frequency Drift with Standard Cuts.....	7
3.2	Reduce Frequency Drift with Minimized Cuts	8
3.3	PLL TX Frequency Drift Measurements	9
4	LR2021 Temperature Compensation.....	10
4.1	Compensation Mechanisms	10
4.2	Implement Frequency Drift Compensation	10
4.2.1	Crystal Frequency Temperature Curve	10
4.2.2	Required Hardware Configuration.....	11
4.2.3	API Commands for Frequency Compensation.....	12
4.3	Performance Results with Compensation	13
4.3.1	Frequency Drift Improvement.....	13
5	Best Practices and Recommendations.....	14
6	Conclusion	15
7	Glossary.....	16
8	Revision History	17

Table Of Figures

Figure 1: TX Frequency Drift During LR2021 Transmission.....5

Figure 2: Example of Thermal Insulation (standard cuts) for LR2021.....7

Figure 3: Wio-LR2021 Module with Minimized cuts (PCB e753v01c)8

Figure 4: Frequency Drift During 22dBm Transmission.....9

Figure 5: NTC Implementation Example 11

Figure 6: TX Frequency Drift Comparison vs Temperature Compensation..... 13

Table Of Tables

Table 1: TX Frequency Drift, Measurement on Two PCB Cuts Dimensions9

1 Introduction

The LR2021 is an advanced ultra-low power transceiver designed to operate across both Sub-GHz and 2.4GHz frequency bands, supporting a variety of modulation schemes including LoRa®, (G)FSK, FLRC, BPSK, OOK, and LR-FHSS. This versatility allows LR2021 to support a variety of wireless protocols, running in the host microcontroller, in addition to LoRaWAN: Bluetooth® Low Energy, WI-SUN, WM-BUS, and Z-Wave.

One of the key challenges when operating a radio transceiver at high power output levels is the thermal effect on frequency stability. This application note addresses the phenomenon of frequency drift in the LR2021 transceiver during high-power transmission, explains its causes, quantifies its effects, and provides recommended mitigation techniques using the built-in temperature compensation capabilities of the LR2021 device.

2 Crystal Thermal Frequency Drift

2.1 Origin of the Frequency Drift

When the LR2021 transmits, especially at high output power, a small but significant amount of heat is generated. This heat is due to the power amplifier (PA) efficiency; a portion of the supplied power is converted to RF signal, while the rest is dissipated as heat.

At +22dBm (158mW) output power, the typical current consumption is approximately 105mA. At 3.3V, this results in a total power consumption of 363mW. After subtracting the 158mW emitted in the RF signal, around 188mW is dissipated as heat in the LR2021.

This heat dissipation affects the 32MHz crystal oscillator (XOSC) in two ways:

- Changes the effective foot capacitance of the on-chip oscillator circuit
- Increases the temperature of the crystal, shifting its resonant frequency

These effects provoke a shift of the radio frequency during the RF packet transmission, which is particularly concerning for long packet durations, as it may cause receive errors if the frequency shifts beyond the receiver's tolerance.

2.2 Impact

The LR2021 is capable of delivering up to +22dBm output power for the sub-GHz band. Measurements show that during a long LoRa® transmission at +22dBm:

- The device and crystal temperatures can increase by several degrees
- Without compensation, the frequency can drift by a few tens of Hertz in the worst cases

This frequency shift is depicted in Figure 1 for an LR2021 continuous emission at 915MHz and at +22dBm output power during a total duration of 45.326 seconds: the frequency drift reaches 90Hz after an emission of 2.2 seconds.

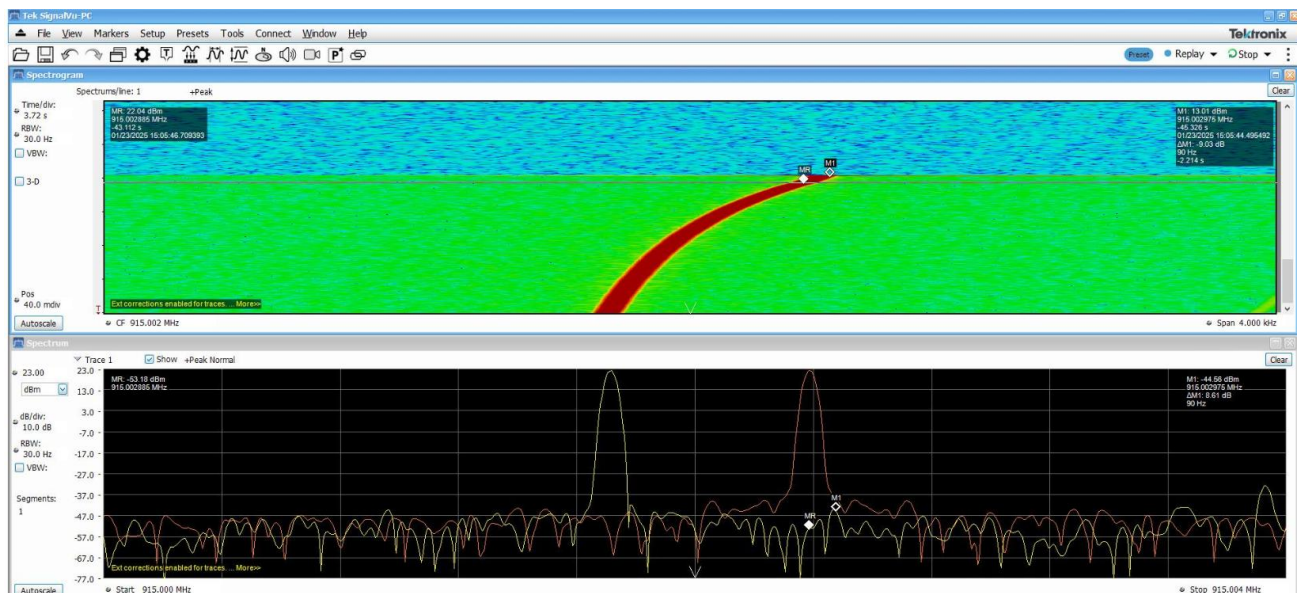


Figure 1: TX Frequency Drift During LR2021 Transmission

The spectrogram display (top plot) shows frequency drift over time, with the red curved trace illustrating the upward frequency drift of the 915MHz carrier signal during continuous transmission. The gradual upward bend demonstrates how the frequency increases over the 45.326-second measurement period, reaching 90Hz drift after just 2.2 seconds of emission.

The spectrum analyzer spectrum (bottom plot) shows the power spectral density across frequency, displaying the main carrier signal frequency offset between the beginning and the end of the transmission, as shown in the spectrogram above.

LoRa® packets using SF12 and BW 125kHz, 10-symbol preamble, and 64-byte payload can last for more than 2 seconds. During this time, the frequency drift may exceed the maximum tolerance for reliable demodulation, potentially causing demodulation failures.

This quantification and consequences of this effect are further described in AN1200.37 dedicated to the SX1261/62 transceivers.

3.3 PLL TX Frequency Drift Measurements

This section shows the frequency drift versus time during a 22dBm continuous transmission (TX CW) at 915MHz.

The measurements were performed on two module PCB designs (18x21.4mm, connected (unsoldered) to the main host MCU board via headers), with the two PCB cuts sizes shown in Figure 2 and Figure 3. Two 32MHz crystals were used: NDK CS06465 (2x1.6mm) and Rakon FTR5238-A (1.6x1.2mm). A third crystal of 1.6x1.2mm size was also measured on the PCB with the small thermal cuts.

The measurement results are summarized in Table 1 and Figure 4 below:

FDRIFT Conditions with TX CW 22dBm 915MHz	Typical (Hz)	Standard Cuts Xtal CS06465 (Hz)	Minimal Cuts Xtal FTR5238-A0 (Hz)	Minimal Cuts Xtal #3 (Hz)
Time freq drift 0.5s	+/- 15.0	12.3	19.4	18.7
Time freq drift 1s	+/- 30.0	22.0	27.8	28.7
Time freq drift 2s	+/- 60.0	54.2	75.3	79.8

Table 1: TX Frequency Drift, Measurement on Two PCB Cuts Dimensions

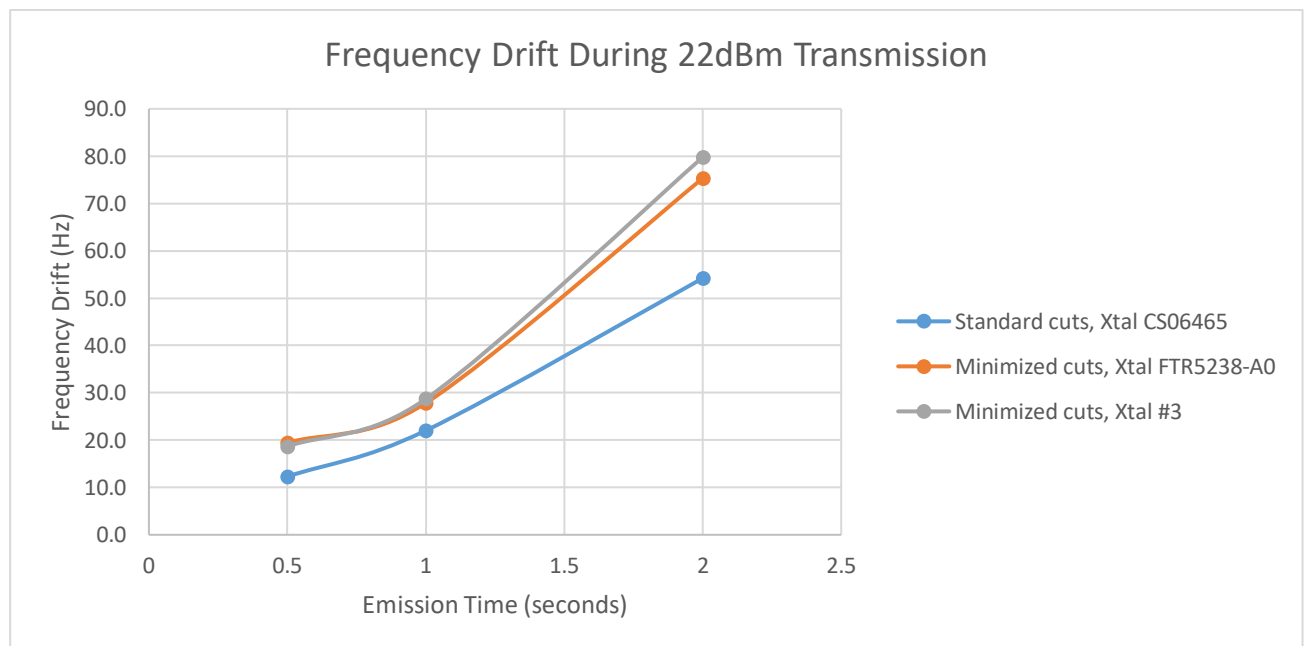


Figure 4: Frequency Drift During 22dBm Transmission

As expected, the frequency drift measured on the PCB with small ground cuts is larger than with standard cuts. Even on the design with small thermal cuts, the frequency shift is around 30Hz/s after one second, which is suitable for successful reception by the RX device (Please refer to Semtech Application Notes AN1200.37 Recommendations for Best Performance (thermal relief) and AN1200.80: LoRa® Modem Doppler Immunity for additional information).

In this very minimal radio setup, the heat generated by the PA does not propagate to a larger PCB, as it would for a module soldered on a main carrier board or on a chip-down design. As such, this test setup represents the worst-case scenario for the thermal drift observable on an LR2021 design.

4 LR2021 Temperature Compensation

The LR2021 incorporates advanced temperature compensation mechanisms to mitigate the thermal frequency drift.

4.1 Compensation Mechanisms

The LR2021 provides two complementary compensation mechanisms:

- **Cfine Compensation:** the LR2021 embeds a temperature sensor located near the XOSC foot capacitors on the chip. This sensor can be used to adjust the fine tuning capacitors (Cfine) to compensate for temperature-dependent changes in the XOSC load capacitance. This Cfine compensation can be activated using command *SetTempCompCfg(...)*
- **XTAL Compensation:** The LR2021 provides an interface for an external Negative Temperature Coefficient (NTC) thermistor to be placed near the crystal on the PCB. This allows for direct monitoring of the crystal's temperature to compensate for the change in the crystal's resonant frequency using a mathematical model of the crystal's S-curve temperature characteristic, based on the external NTC.

4.2 Implement Frequency Drift Compensation

4.2.1 Crystal Frequency Temperature Curve

AT-cut crystals typically have an S-shaped frequency vs. temperature characteristic that can be modeled with a third-order polynomial:

$$\Delta F/F = c_1(T-T_0) + c_2(T-T_0)^2 + c_3(T-T_0)^3$$

Where:

- $\Delta F/F$ is the normalized frequency deviation
- T is the current temperature
- T_0 is the reference temperature (typically 25°C)
- c_1 , c_2 , and c_3 are crystal-specific parameters

The LR2021 uses this model to predict and compensate for the crystal's frequency drift due to the temperature increase during transmission.

4.2.2 Required Hardware Configuration

To enable the frequency compensation via an NTC:

- Connect an NTC with a nominal β of 4250K to the NTC pin through a resistive divider circuit
- Place the NTC as close as possible to the crystal
- Use proper thermal design to minimize heat transfer from the PA to the crystal, adding thermal insulation between the LR2021 and the crystal (cuts in the ground plane on all layers of the PCB)

Refer to **Figure 5** for an NTC implementation example, extracted from Semtech's reference design.

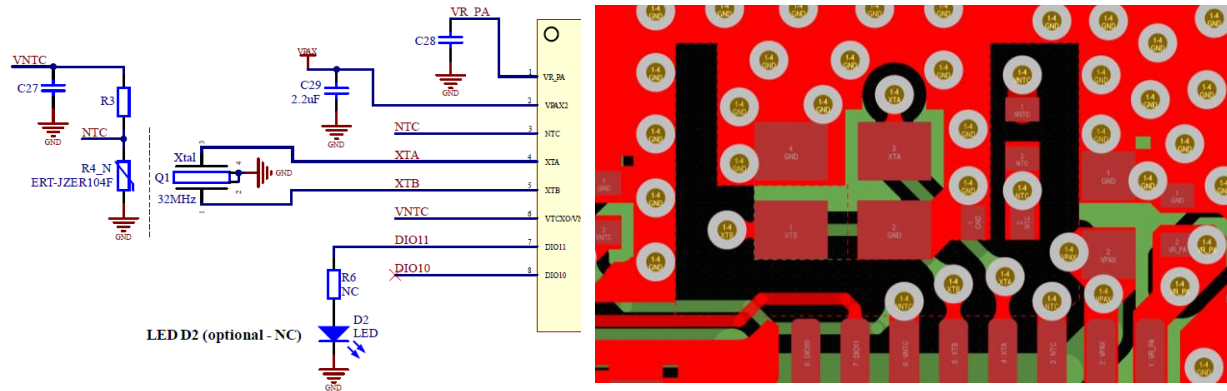


Figure 5: NTC Implementation Example

4.2.3 API Commands for Frequency Compensation

The LR2021 provides several API commands to configure and control frequency drift compensation.

4.2.3.1 SetTempCompCfg

The command *SetTempCompCfg* sets both temperature compensation mechanisms, via Cfine or via the NTC:

- Opcode: 0x0132
- Parameters:
 - Ntc - Enable NTC sensor (1) or disable NTC and enable Cfine (0)
 - CompMode – Activated Compensation mode
 - 0: Disabled
 - 1: Relative mode
 - 2: Absolute mode

Relative mode compensates for the temperature difference between the beginning and the end of the RF packet transmission, which is the preferred approach when it is intended to compensate for the device's self-heating due to packets transmission.

Absolute mode compensates for any difference in temperature measurements. It is therefore more sensitive to any offset in temperature measurement.

Cfine compensation is set when one of the compensation modes is activated and the NTC sensor is disabled.

4.2.3.2 SetNtcParams

The command *SetNtcParams* configures the NTC parameters.

- Opcode: 0x0133
- Parameters:
 - ntc_r_ratio – NTC resistance bias ratio (10-bit value)
 - ntcbeta - NTC beta value (12-bit value)
 - delay - First order filter delay

ntc_r_ratio is the ratio between the bias resistor and NTC resistance at 25°C. It is recommended to set *ntc_r_ratio*=1.

Ntcbeta is the NTC beta coefficient, taken from the NTC specifications (nominal value around 4250 Ohm).

Since the NTC is physically separated from the 32MHz crystal on the PCB, there is a short time interval until the heat generated by the device self-heating gets propagated to the crystal or to the NTC. *Delay* compensates for this time interval and is PCB dependent. The value *Delay*=5, corresponding to 100ms, has been determined as optimal on Semtech's reference design, and is therefore applicable for all exactly similar designs.

4.2.3.3 SetXoscCpTrim

The command *SetXoscCpTrim* can configure the internal loading capacitors to tune the absolute crystal oscillation frequency, in case there is a difference from exactly 32.000MHz due to wrong loading capacitance value. It does not compensate the crystal frequency drift due to the device self heating.

- Opcode: 0x0131
- Parameters:
 - xta - XTA foot capacitor trimming (6-bit value). 0 = 11.3pF, max = 47 = 33.4pF, 1 LSB=0.47pF
 - xtb - XTB foot capacitor trimming (6-bit value). 0 = 11.3pF, max = 47 = 33.4pF, 1 LSB=0.47pF
 - additional_start_time - Additional delay for XTAL start-up.

additional_start_time is an optional parameter which adds an additional delay in us for XTAL starting (RC->XSOC mode) to allow a better stabilization of XTAL. If not set, the previous value is kept. By default *additional_start_time* =0.

4.3 Performance Results with Compensation

4.3.1 Frequency Drift Improvement

The following measurement results show the TX frequency drift during an SF12 BW LoRa packet transmission at 915MHz on the same PCB, for three different conditions: no temperature compensation, Cfine only compensation, and full compensation (Cfine + NTC).

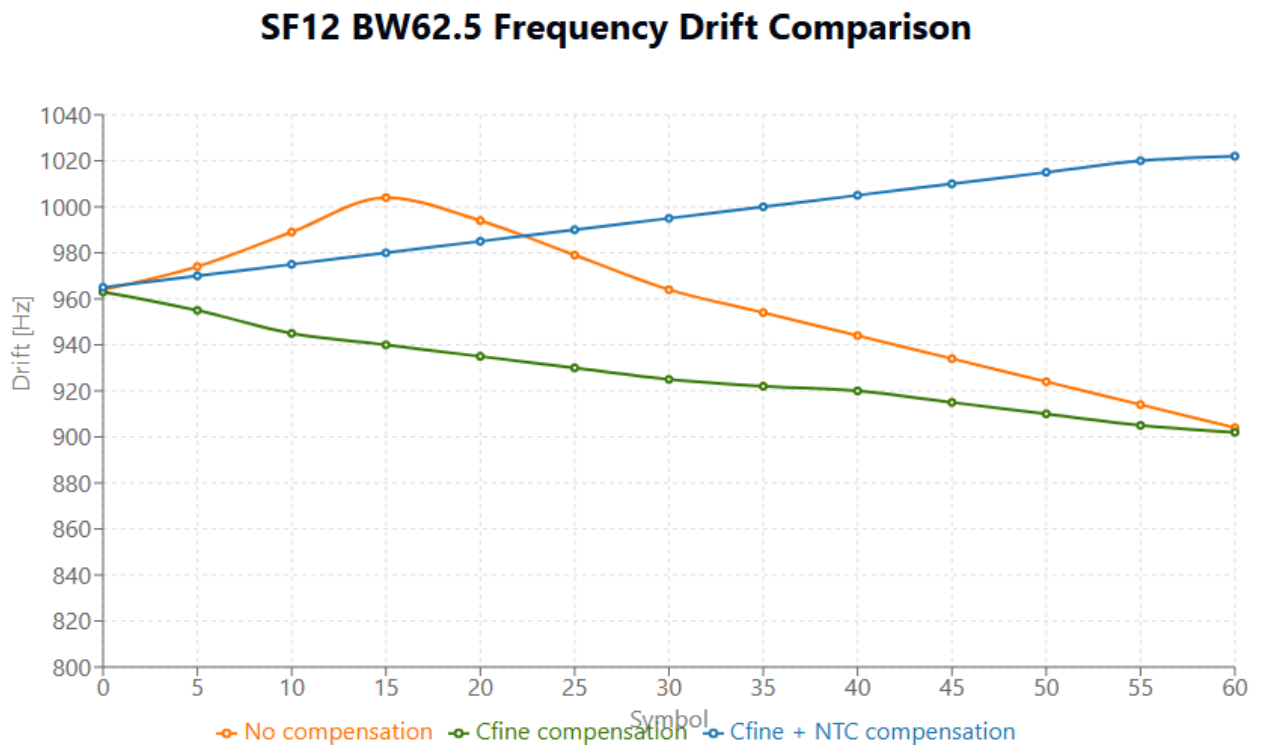


Figure 6: TX Frequency Drift Comparison vs Temperature Compensation

With proper compensation enabled, the following improvements are observed:

- Without Compensation (orange curve), the frequency drift can exceed 100Hz during a long LoRa® packet (SF12, BW 62.5 kHz), potentially resulting in demodulation errors.
- With Cfine Compensation only (green curve), the drift is reduced to 61Hz, but still has the same trend as without compensation, showing that Cfine compensation attenuates the frequency drift but does not compensate it completely.
- With Full Compensation (Cfine + Xtal via NTC, blue curve), the frequency drift is reduced to 57Hz. The upward frequency variation results from an overcompensation mechanism, proving that the NTC compensation is able to correct entirely the frequency drift.

5 Best Practices and Recommendations

Always enable temperature compensation for applications that have high power (>14dBm) and long packet durations.

Use the DC-DC converter configuration when possible, as it reduces overall power consumption and heat generation.

During PCB design, consider thermal insulation between the LR2021 and the crystal, for ground plane cuts and strategic component placement.

Cfine compensation allows to largely reduce the thermal drift, but for applications requiring maximum frequency stability, enable both Cfine and NTC compensation mechanisms simultaneously to obtain the highest thermal correction, typically an 80-90% drift reduction compared to uncompensated operation.

At temperatures below -20°C or above 70°C, the crystal's frequency deviation is more pronounced. In these cases, it is recommended to use a common Temperature Compensated Crystal Oscillator (TCXO). A TCXO is also recommended if the frequency precision requirement is beyond the crystal specifications (e.g. for low bandwidth, or for Round Trip Time of Flight device to device ranging).

6 Conclusion

Frequency drift due to thermal effects is an inherent challenge in high-power RF transmitters like the LR2021. However, with the advanced temperature compensation features available in the device, this effect can be effectively mitigated, ensuring reliable communications even with challenging configurations and long packet durations.

By following the guidelines in this application note and implementing the appropriate compensation mechanisms, designers can achieve optimal performance from the LR2021 transceiver across a wide range of operating conditions.

7 Glossary

Term	Description
API	Application Programming Interface
BW	Bandwidth
Cfine	Fine Tuning Capacitors
DC-DC	Direct Current to Direct Current Converter
FLRC	Fast Long Range Communication
(G) FSK	(Gaussian) Frequency Shift Keying
(G) MSK	(Gaussian) Minimum Shift Keying
LoRa®	Long Range Communication the LoRa® Mark is a registered trademark of the Semtech Corporation
LoRaWAN®	Long Range Wide Area Network
MCU	MicroController Unit
NTC	Negative Temperature Coefficient
OOK	On–Off Keying
OQPSK	Offset Quadrature Phase-Shift Keying
PA	Power Amplifier
PLL	Phase-Locked Loop
PCB	Printed Circuit Board
RF	Radio Frequency
RX	Receiver / Receive
SF	Spreading Factor
TCXO	Temperature Compensated Crystal Oscillator
Tx	Transmitter / Transmit
WM-Bus	Wireless version of M-Bus, which is a European standard (EN 13757) that describes remote meter reading systems
XOSC	Crystal Oscillator
Xtal	Crystal
WI-SUN	Wireless Smart Utility Network

8 Revision History

Version	ECO	Date	Changes and/or Modifications
1.0	076678	August 2025	Initial Release



Important Notice

Information relating to this product and the application or design described herein is believed to be reliable, however such information is provided as a guide only and Semtech assumes no liability for any errors in this document, or for the application or design described herein.

Semtech reserves the right to make changes to the product or this document at any time without notice. Buyers should obtain the latest relevant information before placing orders and should verify that such information is current and complete. Semtech warrants performance of its products to the specifications applicable at the time of sale, and all sales are made in accordance with Semtech's standard terms and conditions of sale.

SEMTECH PRODUCTS ARE NOT DESIGNED, INTENDED, AUTHORIZED OR WARRANTED TO BE SUITABLE FOR USE IN LIFE-SUPPORT APPLICATIONS, DEVICES OR SYSTEMS, OR IN NUCLEAR APPLICATIONS IN WHICH THE FAILURE COULD BE REASONABLY EXPECTED TO RESULT IN PERSONAL INJURY, LOSS OF LIFE OR SEVERE PROPERTY OR ENVIRONMENTAL DAMAGE. INCLUSION OF SEMTECH PRODUCTS IN SUCH APPLICATIONS IS UNDERSTOOD TO BE UNDERTAKEN SOLELY AT THE CUSTOMER'S OWN RISK.

Should a customer purchase or use Semtech products for any such unauthorized application, the customer shall indemnify and hold Semtech and its officers, employees, subsidiaries, affiliates, and distributors harmless against all claims, costs damages and attorney fees which could arise.

The Semtech name and logo are registered trademarks of the Semtech Corporation. All other trademarks and trade names mentioned may be marks and names of Semtech or their respective companies. Semtech reserves the right to make changes to, or discontinue any products described in this document without further notice. Semtech makes no warranty, representation or guarantee, express or implied, regarding the suitability of its products for any particular purpose. All rights reserved.

© Semtech 2025